Problem H2: Electroweak Mixing of $W^{(1,2,3)}$ and B

Worked entirely from Griffiths Ch. 9 §9.7

Parker — PHYS 7502

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Problem Statement

H2. Give the relation that describes the "mixing" of the "fundamental" W_0 (the $\{W_1, W_2, W_3\}$ vector boson triplet) and B vector bosons of the electroweak theory, which get transformed in terms of the weak mixing angle into the usual 4 electroweak bosons we observe (W^+, W^-, Z^0, γ).

Deliverables

1. Give the relation between the charged physical bosons $W^{\pm}_{\mu}$ and the W^1_{μ}, W^2_{μ} triplet members (Griffiths Eq. 9.148).
2. Give the rotation between the neutral physical bosons $\{A_\mu, Z_\mu\}$ and the neutral fundamentals $\{W^3_\mu, B_\mu\}$, in terms of the weak mixing angle θ_W (Griffiths Eq. 9.151).
3. Give the consistency relation $g_W \sin\theta_W = g' \cos\theta_W = g_e$ (Griffiths Eq. 9.154).
4. Give Griffiths' answer to "why does the mixing happen?"

Background & References

Sole reference for this problem.

Griffiths, *Introduction to Elementary Particles*, 2nd edition, Wiley-VCH (2008).

- **Ch. 9 "Weak Interactions"**, specifically:
  - §9.6 "Neutral Weak Interactions" (background — the existence of the Z^0 and Table 9.1 of c_V, c_A values)
  - **§9.7 "Electroweak Unification"** — the heart of this problem, with three subsections:
    - §9.7.1 "Chiral Fermion States" (Eqs. 9.116–9.137)
    - §9.7.2 "Weak Isospin and Hypercharge" (Eqs. 9.134–9.144)
    - **§9.7.3 "Electroweak Mixing"** (Eqs. 9.145–9.158)
- **Ch. 10 "Gauge Theories"**, §10.9 "The Higgs Mechanism". Griffiths defers the dynamical *cause* of the mixing to this section; Part 7 of this writeup gives a one-paragraph summary.

Key Griffiths equations used in this writeup:

Eq. 9.119–9.122:	chiral spinor projectors $(1 \mp \gamma^5)/2$
Eq. 9.134:	LH doublet χ_L
Eq. 9.137:	$\tau^{\pm} = \frac{1}{2}(\sigma^1 \pm i\sigma^2)$
Eq. 9.139:	Gell-Mann–Nishijima $Q = T_3 + Y/2$
Eq. 9.145:	the GWS Lagrangian (the postulate)
Eq. 9.148:	$\boxed{W_\mu^{\pm} = \frac{1}{\sqrt{2}}(W_\mu^1 \pm iW_\mu^2)}$
Eq. 9.151:	$\boxed{A_\mu = B_\mu \cos\theta_W + W_\mu^3 \sin\theta_W, \quad Z_\mu = -B_\mu \sin\theta_W + W_\mu^3 \cos\theta_W}$
Eq. 9.154:	$\boxed{g_W \sin\theta_W = g' \cos\theta_W = g_e}$
Eq. 9.156:	$g_z = g_e/(\sin\theta_W \cos\theta_W)$.

Intuition

Roadmap (entirely along Griffiths' lines). Griffiths develops the GWS theory in *four* stages within §9.7. We follow him in the same order:

1. **§9.7.1 (Stage 1):** Recast the weak vertex factor in terms of *chiral* spinors u_L, u_R. This shows that the weak vertex is purely vectorial in the chiral basis, but couples only to LH states.
2. **§9.7.2 (Stage 2):** Build a left-handed doublet χ_L and recognize an $SU(2)_L$ "weak isospin." Notice it has only $\tau^{\pm}$ pieces (charged currents). Try a τ^3 piece (would-be neutral current) but it doesn't match experiment because real NC has both LH and RH pieces. Fix this by introducing *weak hypercharge* Y via the Gell-Mann–Nishijima formula $Q = T_3 + \frac{Y}{2}$.
3. **§9.7.3 (Stage 3 — the GWS postulate):** The three weak isospin currents couple to a triplet $\vec{W}^\mu$ with strength g_W, the weak hypercharge current couples to a singlet B^μ with strength $g'/2$ (Eq. 9.145). This is the starting point.
4. **§9.7.3 (Stage 4 — the mixing):** Charge sector $(W^1, W^2) \to W^{\pm}$ is forced by the structure of $\tau^{\pm}$. Neutral sector $(W^3, B) \to (\gamma, Z)$ is *stated* (Eq. 9.151) as part of GWS theory; the dynamical reason is the Higgs mechanism, which Griffiths covers later in §10.9.

We'll teach each stage with all the intermediate steps Griffiths skips for compactness, plus extra physical motivation. **Every equation, notation, and result in what follows comes from Griffiths.**

Part 1 — Chiral fermion states (following Griffiths §9.7.1)

Problem Statement

What this part does. Sets the stage by recasting the V-A weak vertex factor in terms of *chiral* spinors u_L and u_R. This makes the weak vertex purely *vectorial* (just like QED), with the V-A structure absorbed into the chirality of the participating spinors. After this rewriting, the weak and electromagnetic interactions will look structurally similar enough to be unified.

Deliverables

1. Define $u_L, u_R, \bar{u}_L, \bar{u}_R$ via the chirality projectors (Griffiths Eqs. 9.119, 9.122, 9.127–9.128).
2. Show how the weak vertex factor $\gamma^\mu(1-\gamma^5)/2$ becomes simply $\bar{\nu}_L\gamma^\mu e_L$ (Griffiths Eq. 9.129).
3. Show how the QED current decomposes in chiral form into purely vectorial pieces for both L and R (Griffiths Eq. 9.135).

1.1 — Define chiral spinors (Griffiths Eqs. 9.119–9.128)

Step 1.1: The four chiral combinations

Griffiths defines (his Eqs. 9.119–9.122 and 9.125–9.126):

$$\boxed{u_L \equiv \frac{(1-\gamma^5)}{2}u(p), \qquad u_R \equiv \frac{(1+\gamma^5)}{2}u(p),}$$

and for antiparticle spinors (the γ^5 sign flips):

$$v_L \equiv \frac{(1+\gamma^5)}{2}v(p), \qquad v_R \equiv \frac{(1-\gamma^5)}{2}v(p).$$

The subscript "L" is meant to suggest "left-handed" (helicity -1 in the massless limit), but as Griffiths emphasizes in his footnote, u_L is only a true helicity eigenstate when $E \gg mc^2$. The terminology is conventional, not literal.

Step 1.2: Adjoint chiral spinors get the opposite projector

Griffiths Eqs. 9.127–9.128. Because γ^5 is Hermitian and anticommutes with γ^μ,

$$\bar{u}_L = u_L^\dagger\gamma^0 = \bar{u}(p)\frac{(1+\gamma^5)}{2}, \qquad \bar{u}_R = \bar{u}(p)\frac{(1-\gamma^5)}{2}.$$

The projector flips for the adjoint. This is the key fact that lets Griffiths absorb the $(1-\gamma^5)$ factor of the V-A vertex into the spinors themselves.

Term Refresher

Why $\bar{u}_L$ has $(1+\gamma^5)/2$ instead of $(1-\gamma^5)/2$:

$$\bar{u}_L \;=\; u_L^\dagger \gamma^0 \;=\; \left[\frac{(1-\gamma^5)}{2}u\right]^\dagger \gamma^0 \;=\; u^\dagger \frac{(1-\gamma^5)}{2}\gamma^0 \;=\; u^\dagger \gamma^0 \frac{(1+\gamma^5)}{2} \;=\; \bar{u}\,\frac{(1+\gamma^5)}{2},$$

using $\gamma^5\gamma^0 = -\gamma^0\gamma^5$. This sign flip is essential.

1.2 — Recast the weak vertex in chiral form (Griffiths Eq. 9.129)

Step 1.3: The vertex factor for $e^- \to \nu_e + W^-$

The V-A weak vertex factor that Griffiths starts from way back at Eq. 9.5 is

$$-i\,\frac{g_W}{\sqrt{2}}\,\gamma^\mu \frac{(1-\gamma^5)}{2}.$$

The matrix element contribution from this vertex is (Griffiths' notation, with e and ν standing for the particle spinors):

$$j_\mu^- \;\equiv\; \bar{\nu}\,\gamma_\mu \frac{(1-\gamma^5)}{2}\,e.$$

This is the "negatively charged weak current" (Griffiths Eq. 9.125 / 9.129). Now use the projector identity (Griffiths Eqs. 9.130–9.132):

$$\gamma^\mu \frac{(1-\gamma^5)}{2} \;=\; \frac{(1+\gamma^5)}{2}\gamma^\mu \;=\; \frac{(1+\gamma^5)}{2}\gamma^\mu \frac{(1-\gamma^5)}{2} \quad \text{(idempotent on the right)}.$$

So

$$\bar{\nu}\,\gamma_\mu \frac{(1-\gamma^5)}{2}\,e \;=\; \bar{\nu}\frac{(1+\gamma^5)}{2}\gamma_\mu \frac{(1-\gamma^5)}{2}\,e \;=\; \bar{\nu}_L \gamma_\mu e_L.$$

Result (Griffiths Eq. 9.129):

$$\boxed{j_\mu^- \;=\; \bar{\nu}_L\,\gamma_\mu\,e_L.}$$

The V-A factor is gone — absorbed into the chirality of the spinors. The vertex factor in this notation is now *purely vectorial* (no γ^5), and it tells you that W^- couples a left-handed electron to a left-handed neutrino.

1.3 — Recast QED in chiral form (Griffiths Eq. 9.135)

Step 1.4: The electromagnetic current in chiral form

The QED interaction has the vertex $-ig_e\gamma^\mu$, and the electromagnetic current of an electron is $j^{\text{em}}_\mu = -\bar{e}\gamma_\mu e$ (the minus sign accounts for the electron's negative charge). Decomposing $e = e_L + e_R$ and using the orthogonality of projectors $P_L P_R = 0$ (Griffiths Eq. 9.137), the cross-terms vanish:

$$\bar{e}\gamma_\mu e \;=\; (\bar{e}_L + \bar{e}_R)\,\gamma_\mu\,(e_L + e_R) \;=\; \bar{e}_L\gamma_\mu e_L + \bar{e}_R\gamma_\mu e_R \quad \text{(cross-terms vanish)}.$$

So

$$\boxed{j^{\text{em}}_\mu \;=\; -\,\bar{e}_L\gamma_\mu e_L \;-\; \bar{e}_R\gamma_\mu e_R.} \quad \text{(Griffiths Eq. 9.135)}$$

The QED current splits into a sum of two purely vectorial pieces, one for each chirality.

Griffiths quote

"Equations (10.133) and (10.135) are beginning to look like the stuff of which one might build a unified theory. It is true that the weak current only couples left-handed states, whereas the electromagnetic current couples both types, but apart from that they are strikingly similar."
— Griffiths, p. 333–334 (1st ed.) / §9.7.1 (2nd ed.)

Physical Insight

Why this rewriting matters. After Step 1.4, both the weak and the electromagnetic currents are written as purely vectorial bilinears of *chiral* spinors. The weak current uses LH only; the EM current uses both LH and RH. **This structural similarity is what makes electroweak unification mathematically possible** — both currents now "live" in the same vector-current language.
In Griffiths' words (paraphrased from his own framing): the V-A factor $(1-\gamma^5)/2$ is not really a "different" kind of vertex; it just selects which chiralities of fermion participate. Once you absorb the projector into the spinor, all gauge interactions look the same — vector currents coupling to gauge fields.

Key Result

Part 1 results (entirely from Griffiths §9.7.1):

- Chiral spinors u_L, u_R are defined by projectors $(1 \mp \gamma^5)/2$ (Eq. 9.119, 9.126).
- The weak charged current becomes $j^-_\mu = \bar{\nu}_L\gamma_\mu e_L$ (Eq. 9.129) — purely vectorial, LH only.
- The EM current becomes $j^{\text{em}}_\mu = -\bar{e}_L\gamma_\mu e_L - \bar{e}_R\gamma_\mu e_R$ (Eq. 9.135) — purely vectorial, both chiralities.

Part 2 — Weak isospin and hypercharge (following Griffiths §9.7.2)

Problem Statement

What this part does. Promotes the LH spinors ν_L, e_L into a doublet, defines weak isospin and weak hypercharge as $SU(2)_L$ and $U(1)$ quantum numbers, and identifies the symmetry group $SU(2)_L \otimes U(1)$ that the GWS Lagrangian is built on. This part introduces the four currents that will couple to the four fundamental gauge fields in Part 3.

Deliverables

1. Build the LH doublet $\chi_L = \binom{\nu_L}{e_L}$ (Griffiths Eq. 9.134).
2. Define the $\tau^{\pm}$ matrices and recognize the $SU(2)$ structure (Griffiths Eq. 9.137).
3. Hint at a third weak current $j^3 \propto \tau^3$ and explain why it is *not yet* the physical neutral current (Griffiths' "Not so fast" moment).
4. Introduce weak hypercharge Y via $Q = T_3 + Y/2$ (Griffiths Eq. 9.139).
5. State the underlying symmetry group $SU(2)_L \otimes U(1)$ and list the four currents (three isospin + one hypercharge).

2.1 — The LH doublet (Griffiths Eq. 9.134)

Step 2.1: Pack the LH spinors into a doublet

Following Griffiths' Eq. 9.134, define the left-handed doublet for the first lepton generation:

$$\boxed{\chi_L \equiv \begin{pmatrix} \nu_L \\ e_L \end{pmatrix}.}$$

The right-handed states e_R, ν_R are *singlets* under the new $SU(2)$ Griffiths is about to introduce — they don't sit in any doublet.
This is structurally analogous to the strong-isospin doublet $\binom{p}{n}$ (Griffiths §4.5), but Griffiths is emphatic in his footnote that weak isospin is *not* the same as strong isospin — they share only mathematical structure (and the unfortunate name).

Term Refresher

Reminder: why does the LH doublet have ν on top and e on bottom? By convention, the upper component carries $T_3 = +\frac{1}{2}$ and the lower carries $T_3 = -\frac{1}{2}$. Looking ahead to the Gell-Mann–Nishijima formula $Q = T_3 + Y/2$, since $Y_{e_L} = Y_{\nu_L}$ (both members of the same doublet share Y), the difference $Q_\nu - Q_e = T_3^{(\nu)} - T_3^{(e)} = +1/2 - (-1/2) = 1$, which matches the observed charge difference $0 - (-1) = 1$.

2.2 — Charged currents in matrix notation (Griffiths Eq. 9.137)

Step 2.2: Step-up and step-down operators

Griffiths Eq. 9.135–9.137. Define the 2×2 matrices (note Griffiths uses τ instead of σ to avoid confusion with ordinary spin):

$$\tau^+ = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}, \qquad \tau^- = \begin{pmatrix} 0 & 0 \\ 1 & 0 \end{pmatrix}.$$

These are linear combinations of the first two Pauli matrices:

$$\boxed{\tau^\pm = \tfrac{1}{2}(\sigma^1 \pm i\,\sigma^2)} \quad \text{(Griffiths Eq. 9.137).}$$

The two charged currents from Step 1.3 can now be packaged together:

$$j_\mu^+ = \bar{\chi}_L\,\gamma_\mu\,\tau^+\,\chi_L = \bar{\nu}_L \gamma_\mu e_L, \qquad j_\mu^- = \bar{\chi}_L\,\gamma_\mu\,\tau^-\,\chi_L = \bar{e}_L \gamma_\mu \nu_L.$$

The matrix algebra is straightforward: e.g., $\tau^+\chi_L = \binom{e_L}{0}$, so $\bar{\chi}_L\gamma_\mu\tau^+\chi_L = (\bar{\nu}_L\ \bar{e}_L)\gamma_\mu\binom{e_L}{0} = \bar{\nu}_L\gamma_\mu e_L$.

2.3 — A would-be neutral current (Griffiths' "Not so fast" moment)

Step 2.3: Try a third matrix τ^3

The natural third matrix in the $\sigma^1, \sigma^2, \sigma^3$ family is

$$\tau^3 = \tfrac{1}{2}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \quad \text{(Griffiths Eq. 9.138).}$$

The corresponding "third weak current" is

$$j_\mu^3 = \bar{\chi}_L\,\gamma_\mu\,\tau^3\,\chi_L = \tfrac{1}{2}\bar{\nu}_L\gamma_\mu\nu_L - \tfrac{1}{2}\bar{e}_L\gamma_\mu e_L \quad \text{(Griffiths Eq. 9.142).}$$

Griffiths quote

"Perfect!" (I hear you exclaim.) "There's the neutral weak current!" Not so fast. This current only couples *left*-handed particles; in the older language it is pure V-A, whereas the neutral weak interaction involves right-handed components as well.
— Griffiths §9.7.2

Physical Insight

So what does Griffiths do? The weak NC *cannot* be just j^3. But j^3 is the natural third member of the $SU(2)_L$ multiplet. So Griffiths' move is: **keep j^3 as it is, and find a second neutral current that does include RH parts**. The two together will mix to give the physical γ (which couples LH+RH equally) and Z^0 (which couples LH+RH unequally, with weights determined by θ_W). The second neutral current is the weak hypercharge current — which is what Griffiths introduces next.

2.4 — Weak hypercharge (Griffiths Eq. 9.139)

Step 2.4: Define Y via Gell-Mann-Nishijima

Griffiths borrows the structure of strong-isospin hypercharge (his Eq. 4.37) and applies it to the *weak* version. Define weak hypercharge Y by the Gell-Mann–Nishijima formula:

$$\boxed{Q \;=\; T_3 \;+\; \tfrac{1}{2}\,Y} \quad \text{(Griffiths Eq. 9.139).}$$

where Q is electric charge (in units of e), T_3 is the third component of weak isospin, and Y is the new conserved quantum number we're introducing. **The hypercharge assignments are chosen so that Q comes out right after rotation by θ_W.** For the lepton doublet (with $T_3 = +1/2$ for ν_L, $T_3 = -1/2$ for e_L, $Q_\nu = 0$, $Q_e = -1$):

$$Y_{\nu_L} = Y_{e_L} = -1 \quad \text{(both members of the LH doublet share the same } Y\text{)}.$$

For the right-handed singlets ($T_3 = 0$):

$$Y_{e_R} = -2, \quad Y_{\nu_R} = 0 \quad \text{(if } \nu_R \text{ exists at all)}.$$

Step 2.5: The weak hypercharge current

Griffiths Eq. 9.143. The hypercharge current is

$$j^Y_\mu \;=\; 2\,j^{\text{em}}_\mu - 2\,j^3_\mu \quad \text{(rewriting Gell-Mann-Nishijima)}.$$

Equivalently, summed over LH and RH parts:

$$j^Y_\mu \;=\; \sum_{f,\chi} Y^{(f,\chi)}\,\bar{\psi}^{(f,\chi)}_\mu \gamma_\mu \psi^{(f,\chi)} \quad \text{(both LH doublet and RH singlets contribute)}.$$

This is the second neutral current we needed. It involves both LH and RH parts.

2.5 — The symmetry group

Concept / Definition

The electroweak gauge group, à la Griffiths (§9.7.2):

$$\boxed{SU(2)_L \otimes U(1)}$$

- "$SU(2)_L$ refers to the weak isospin (with a subscript to remind us that it involves left-handed states only)." (Griffiths)
- "$U(1)$ refers to weak hypercharge (involving both chiralities)."

Note: Griffiths writes $U(1)$ without a subscript Y in §9.7.2, but it is the same as the $U(1)_Y$ used by other authors. We'll just write $U(1)$ to match Griffiths.

2.6 — The four currents

Key Result

Summary of currents (Griffiths Eqs. 9.142–9.144):
Three weak isospin currents (couple to the $SU(2)_L$ triplet $\vec{W}_\mu$):

$$j^a_\mu \;=\; \bar{\chi}_L\,\gamma_\mu\,\tau^a\,\chi_L, \qquad a = 1, 2, 3.$$

One weak hypercharge current (couples to the $U(1)$ singlet B_μ):

$$j^Y_\mu \;=\; 2\,j^{\text{em}}_\mu - 2\,j^3_\mu \quad \text{(Griffiths Eq. 9.143)}.$$

These four currents will be the inputs to the GWS Lagrangian in Part 3.

Part 3 — The GWS postulate (Griffiths Eq. 9.145)

Problem Statement

What this part does. States the GWS Lagrangian as a postulate (Griffiths Eq. 9.145). This is where the four "fundamental" bosons $\{W^1, W^2, W^3, B\}$ enter the picture.

Griffiths quote

"Now, the GWS model asserts that the three weak isospin currents couple, with strength g_W, to a weak isotriplet of vector bosons $\vec{W}$, whereas the weak hypercharge current couples with strength $g'/2$ to an isosinglet B:"

$$\mathcal{L} \;=\; -i\Big[g_W\,\vec{j}_\mu \cdot \vec{W}^\mu \;+\; \tfrac{g'}{2}\,j^Y_\mu B^\mu\Big] \qquad \text{(Eq. 9.145)}.$$

"(These four particles correspond, ultimately, to the weak and electromagnetic mediators: $W^\pm, Z^0$, and γ — but with a twist, as we shall soon see.)"
— Griffiths §9.7.3

Concept / Definition

Reading Griffiths' Eq. 9.145.

- $\vec{j}_\mu = (j^1_\mu, j^2_\mu, j^3_\mu)$ is a 3-vector in weak isospin space (the *vector arrow* denotes weak-isospin space, not real space).
- $\vec{W}^\mu = (W^{1\mu}, W^{2\mu}, W^{3\mu})$ is the corresponding 3-vector of gauge fields.
- Their dot product is $\vec{j}\cdot\vec{W} = j^1W^1 + j^2W^2 + j^3W^3$ (Griffiths Eq. 9.146).
- j^Y is the weak hypercharge current; B^μ is the corresponding singlet gauge field.
- There are *two* independent coupling constants: g_W (for the $SU(2)_L$ triplet) and g' (for the $U(1)$ singlet, with the conventional factor $\frac{1}{2}$ absorbed into the Lagrangian to match the $Y/2$ in the Gell-Mann-Nishijima formula).

Intuition

Why exactly 4 fundamental gauge fields? Griffiths doesn't dwell on the counting, but you should: $SU(2)_L$ has 3 generators (τ^1, τ^2, τ^3, or equivalently $\sigma^1/2, \sigma^2/2, \sigma^3/2$), and $U(1)$ has 1. Each generator of a gauge symmetry gives rise to one gauge field, so there are $3+1 = 4$ fundamental bosons. After the mixing in Part 5, these four become the four *physical* bosons $W^\pm, Z^0, \gamma$.

Key Result

The GWS Lagrangian (the starting point):

$$\boxed{\mathcal{L}^{\text{GWS}}_{\text{int}} \;=\; -i\Big[g_W\,\vec{j}_\mu \cdot \vec{W}^\mu \;+\; \tfrac{g'}{2}\,j^Y_\mu B^\mu\Big]} \qquad \text{(Griffiths Eq. 9.145)}.$$

Everything else in this problem is consequence of this Lagrangian plus the assumption that the symmetry $SU(2)_L \otimes U(1)$ is broken to $U(1)_{\text{em}}$ at low energies.

Part 4 — Charged sector $W^{\pm} = \frac{1}{\sqrt{2}}(W^1 \pm iW^2)$

Problem Statement

What this part does. Take the charged piece $j^1W^1 + j^2W^2$ of Eq. 9.145 and rewrite it in terms of the physical charged bosons $W^{\pm}$. **This is just algebra on the GWS Lagrangian; no mixing angle is involved.**

Deliverables

1. Write $\vec{j} \cdot \vec{W}$ in terms of charged currents $j^{\pm} = j^1 \pm ij^2$ and physical charged fields $W^{\pm}$ (Griffiths Eq. 9.147).
2. Read off the definition $W^{\pm}_{\mu} = \frac{1}{\sqrt{2}}(W^1_{\mu} \pm iW^2_{\mu})$ (Griffiths Eq. 9.148).
3. Verify that this gives back the original V-A vertex factor for $e^- \to \nu_e + W^-$ (Griffiths Eqs. 9.149–9.150).

4.1 — Rewrite $\vec{j} \cdot \vec{W}$ using charged currents

Step 4.1: Charged currents combine W^1 and W^2

Define the charged currents (Griffiths Eq. 9.147):

$$j^{\pm}_{\mu} \equiv j^1_{\mu} \pm i\, j^2_{\mu}.$$

We want to rewrite $j^1W^1 + j^2W^2$ in terms of $j^{\pm}$ and some new fields $W^{\pm}$. The trick is to note that

$$j^1W^1 + j^2W^2 = \tfrac{1}{2}\left[(j^1 + ij^2)(W^1 - iW^2) + (j^1 - ij^2)(W^1 + iW^2)\right].$$

Verification (one line of algebra). Expand the right side:

$$\begin{aligned}
&\tfrac{1}{2}\left[(j^1 + ij^2)(W^1 - iW^2) + (j^1 - ij^2)(W^1 + iW^2)\right] \\
&\quad = \tfrac{1}{2}\left[j^1W^1 - ij^1W^2 + ij^2W^1 + j^2W^2 \;+\; j^1W^1 + ij^1W^2 - ij^2W^1 + j^2W^2\right] \\
&\quad = \tfrac{1}{2}\left[2j^1W^1 + 2j^2W^2\right] \;=\; j^1W^1 + j^2W^2. \quad \checkmark
\end{aligned}$$

The cross-terms $\pm ij^1W^2$ and $\mp ij^2W^1$ cancel exactly. **This is the whole point of using complex combinations.**

Step 4.2: Identify $W^\pm$ from this rewriting

Setting $j^\pm = j^1 \pm i j^2$ on the current side, we read off the corresponding gauge field combinations on the boson side. To get the $\frac{1}{\sqrt{2}}$ normalizations that match the standard V-A vertex factor (Griffiths' next move), define

$$\boxed{W_\mu^\pm \equiv \frac{1}{\sqrt{2}}\left(W_\mu^1 \pm i\, W_\mu^2\right) \quad \textbf{(Griffiths Eq. 9.148)}}$$

Then

$$j^1 W^1 + j^2 W^2 = \tfrac{1}{\sqrt{2}}\left(j^+ W^- + j^- W^+\right) \quad \text{(Griffiths Eq. 9.147 second line).}$$

Intuition

Where the $\frac{1}{\sqrt{2}}$ comes from. You can see it most cleanly by demanding that the kinetic term of the gauge fields stay canonically normalized. In the original basis $-\frac{1}{4}[(F^1_{\mu\nu})^2+(F^2_{\mu\nu})^2]$ with two real fields. Substituting $W^1 = \frac{1}{\sqrt{2}}(W^+ + W^-)$ and $W^2 = \frac{i}{\sqrt{2}}(W^+ - W^-)$ gives $-\frac{1}{2}F^+_{\mu\nu}F^{-\,\mu\nu}$ with the conventional factor of $\frac{1}{2}$ for a complex (charged) vector field. The $\sqrt{2}$ is bookkeeping that keeps everyone canonically normalized.

4.2 — Verify the V-A vertex factor (Griffiths Eqs. 9.149–9.150)

Step 4.3: Pluck out the W^- vertex

The charged piece of the GWS Lagrangian (Griffiths Eq. 9.145, restricted to charged sector) is, using Step 4.2:

$$\mathcal{L}_{\text{CC}} = -i g_W \cdot \tfrac{1}{\sqrt{2}}\left(j^+ W^- + j^- W^+\right).$$

For the process $e^- \to \nu_e + W^-$, we use the $j^- W^+$ piece... wait, let me be more careful: this process has a W^- in the final state, so it couples to the field that *creates* a W^-, which is $W^{-\,\dagger} = W^+$ acting from the right, or equivalently the $j^- W^+$ term. But following Griffiths' Eq. 9.149 conventions, we look at the $j^- W^+$ piece:

$$\mathcal{L} \supset -i\frac{g_W}{\sqrt{2}}\, j^-_\mu\, W^{+\mu} = -i\frac{g_W}{\sqrt{2}}\,\bar{e}_L \gamma_\mu \nu_L\, W^{+\mu} = -i\frac{g_W}{\sqrt{2}}\,\bar{e}\gamma_\mu \tfrac{(1-\gamma^5)}{2}\nu\, W^{+\mu}.$$

The vertex factor for the $\bar{e}\nu W^+$ vertex is therefore (Griffiths Eq. 9.150):

$$\boxed{-i\,\frac{g_W}{\sqrt{2}}\,\gamma^\mu\frac{(1-\gamma^5)}{2} \quad \text{(Griffiths Eq. 9.150)}}$$

Which is exactly what we started with (Griffiths' very first weak vertex factor, Eq. 9.5). The construction is self-consistent.

Griffiths quote

"...which is exactly what we started with (Equation 9.5)." — Griffiths §9.7.3

4.3 — Convention warning

Warning / Convention Pitfall

Sign convention. Griffiths uses

$$W_\mu^\pm = \tfrac{1}{\sqrt{2}}(W_\mu^1 \pm iW_\mu^2) \quad \text{(Eq. 9.148)}.$$

Many other textbooks (Thomson, Mann, Halzen & Martin) use the *opposite* sign: $W^\pm = \frac{1}{\sqrt{2}}(W^1 \mp iW^2)$. This is just a labeling convention — it flips which combination is called W^+ versus W^-. The physics is identical, and as long as you're consistent within a single textbook, the V-A vertex factor comes out the same. **For Frantz's class, follow Griffiths.**

Key Result

Part 4 result (Griffiths Eq. 9.148):

$$\boxed{W_\mu^\pm \;=\; \frac{1}{\sqrt{2}}\left(W_\mu^1 \pm i\, W_\mu^2\right).}$$

This is a unitary change of basis: two real fields W^1, W^2 get repackaged as a complex field W^+ and its conjugate W^-. **No mixing angle, no Higgs, no θ_W.** It's just $SU(2)$ raising/lowering structure.

Part 5 — Neutral mixing $\{W^3, B\} \to \{\gamma, Z\}$ (Griffiths Eq. 9.151)

Problem Statement

What this part does. *States* the neutral mixing relation as part of the GWS theory — this is the central deliverable for H2. Griffiths himself does not derive this from a more fundamental principle in §9.7.3; he asserts it (with the dynamical mechanism deferred to §10.9, the Higgs mechanism). We follow him in the same way.

Deliverables

1. State the mixing relation between $\{W^3, B\}$ and $\{\gamma, Z\}$ as Griffiths does in Eq. 9.151.
2. Briefly explain why $W^3 \neq \gamma$ (so a true mixing is needed).
3. Note that Griffiths defers the dynamical reason for the mixing to §10.9 (Higgs mechanism); we'll do the same and pick that up in Part 7.

5.1 — Why neither W^3 alone nor B alone can be the photon

Step 5.1: W^3 couples LH only — so $W^3 \neq \gamma$

From Step 2.3 we have $j^3_\mu = \frac{1}{2}(\bar{\nu}_L \gamma_\mu \nu_L - \bar{e}_L \gamma_\mu e_L)$, which involves *only* left-handed states. The photon, on the other hand, couples to both LH and RH states (Step 1.4). **Therefore W^3 alone cannot be the photon**, or QED would be parity-violating, contradicting experiment.

Step 5.2: B couples to Y

The B-boson couples to j^Y, the weak hypercharge current. But hypercharge is *not* electric charge: e.g., the neutrino has $Q_\nu = 0$ but $Y_{\nu_L} = -1$. So if B were the photon, the neutrino would have a non-zero electromagnetic coupling — which it doesn't. **Therefore B alone is not the photon either.**

Physical Insight

So the photon must be a mixture. The Gell-Mann–Nishijima formula $Q = T_3 + Y/2$ tells us that electric charge is a linear combination of weak isospin (T_3) and weak hypercharge (Y). Since W^3 couples to T_3 and B couples to Y, the gauge field that couples to Q must be a linear combination of W^3 and B. The orthogonal combination is the Z^0.

5.2 — The mixing relation (Griffiths Eq. 9.151)

Griffiths quote

"But the underlying $SU(2)_L \otimes U(1)$ symmetry is 'broken,' in GWS theory: the two neutral states, W^3 and B, 'mix,' producing one massless linear combination (the photon), and an orthogonal massive combination (the Z^0):"

$$\begin{aligned} A_\mu &= B_\mu \cos\theta_W + W^3_\mu \sin\theta_W \\ Z_\mu &= -B_\mu \sin\theta_W + W^3_\mu \cos\theta_W \end{aligned} \qquad \text{(Eq. 9.151)}$$

"(You see now why θ_W is called the 'weak mixing angle.')"
— Griffiths §9.7.3

Key Result

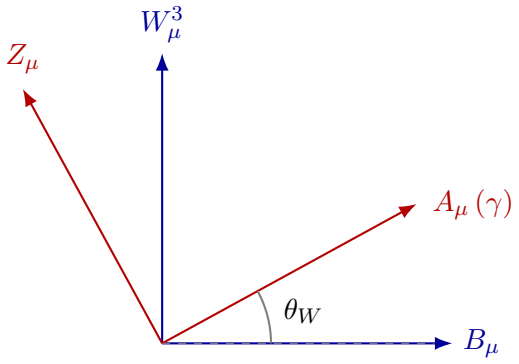
The neutral mixing (Griffiths Eq. 9.151), the central deliverable for H2:

$$\boxed{\begin{aligned} A_\mu &= B_\mu \cos\theta_W + W^3_\mu \sin\theta_W \quad (=\gamma), \\ Z_\mu &= -B_\mu \sin\theta_W + W^3_\mu \cos\theta_W . \end{aligned}}$$

5.3 — The inverse relation

We will need the inverse for Part 6. The matrix in Eq. 9.151 is $R = \left(\begin{smallmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{smallmatrix}\right)$, an orthogonal matrix with $R^{-1} = R^T$. Inverting:

$$\begin{pmatrix} B_\mu \\ W^3_\mu \end{pmatrix} = \begin{pmatrix} \cos\theta_W & -\sin\theta_W \\ \sin\theta_W & \cos\theta_W \end{pmatrix} \begin{pmatrix} A_\mu \\ Z_\mu \end{pmatrix}, \qquad \boxed{\begin{aligned} B_\mu &= A_\mu \cos\theta_W - Z_\mu \sin\theta_W , \\ W^3_\mu &= A_\mu \sin\theta_W + Z_\mu \cos\theta_W . \end{aligned}} \tag{1}$$



Griffiths Eq. 9.151 as a rotation in the neutral-field plane. Photon A and Z are obtained from B and W^3 by rotation through θ_W.

Part 6 — Determining θ_W from QED matching (Griffiths Eq. 9.154)

Problem Statement

What this part does. Substitute the inverse rotation (1) back into the GWS Lagrangian, and demand that the photon coupling be the standard QED coupling $-ig_e j^{\text{em}} A^\mu$. Out drops Griffiths' Eq. 9.154: the consistency relation among g_W, g', and g_e.

Deliverables

1. Substitute Eq. (1) into the neutral piece of Eq. 9.145 to get the Lagrangian in terms of A_μ and Z_μ (Griffiths Eq. 9.152).
2. Read off the photon coupling and compare to QED (Griffiths Eq. 9.153).
3. Use Gell-Mann–Nishijima at the current level ($j^{\text{em}} = j^3 + \frac{1}{2} j^Y$) to match coefficients.
4. Out drops $g_W \sin\theta_W = g' \cos\theta_W = g_e$ (Griffiths Eq. 9.154).

6.1 — Substitute into the Lagrangian (Griffiths Eq. 9.152)

Step 6.1: Plug W^3 and B in terms of A and Z

The neutral piece of Griffiths Eq. 9.145 is

$$\mathcal{L}_{\text{neut}} \;=\; -\,i\Big[g_W\, j^3_\mu W^{3\mu} \;+\; \tfrac{g'}{2}\, j^Y_\mu B^\mu\Big]\,.$$

Use the inverse rotation (1) and substitute:

$$\begin{aligned}
\mathcal{L}_{\text{neut}} &= -\,i\Big[g_W\, j^3_\mu (A^\mu \sin\theta_W + Z^\mu \cos\theta_W) \;+\; \tfrac{g'}{2}\, j^Y_\mu (A^\mu \cos\theta_W - Z^\mu \sin\theta_W)\Big] \\
&= -\,i\Big[\, A^\mu \Big(g_W \sin\theta_W\, j^3_\mu \;+\; \tfrac{g'}{2} \cos\theta_W\, j^Y_\mu\Big) \\
&\qquad\qquad + Z^\mu \Big(g_W \cos\theta_W\, j^3_\mu \;-\; \tfrac{g'}{2} \sin\theta_W\, j^Y_\mu\Big)\Big]\,.
\end{aligned}$$

This matches Griffiths Eq. 9.152.

6.2 — The QED constraint (Griffiths Eq. 9.153)

Step 6.2: Demand QED form for the photon coupling

Griffiths' Eq. 9.153 says the QED interaction Lagrangian is

$$\mathcal{L}_{\text{QED}} \;=\; -\,i\, g_e\, j^{\text{em}}_\mu A^\mu.$$

For the GWS theory to reproduce QED at low energies, the A^μ piece in our rotated Lagrangian (Step 6.1) must equal $-ig_e j^{\text{em}}_\mu A^\mu$:

$$g_W \sin\theta_W\, j^3_\mu \;+\; \tfrac{g'}{2} \cos\theta_W\, j^Y_\mu \;\stackrel{!}{=}\; g_e\, j^{\text{em}}_\mu.$$

6.3 — Match coefficients using Gell-Mann-Nishijima

Step 6.3: Decompose j^{em} and match

Griffiths uses Eq. 9.143 ($j^Y = 2j^{\text{em}} - 2j^3$, equivalently $j^{\text{em}} = j^3 + \frac{1}{2}j^Y$). Substituting:

$$g_W \sin\theta_W \, j^3_\mu \; + \; \tfrac{g'}{2} \cos\theta_W \, j^Y_\mu \;=\; g_e\big(j^3_\mu \; + \; \tfrac{1}{2}\, j^Y_\mu\big)\,.$$

Match coefficients of j^3_μ and j^Y_μ *separately* (since they're independent currents):

$$\text{coeff. of } j^3: \quad g_W \sin\theta_W \;=\; g_e,$$
$$\text{coeff. of } j^Y: \quad \tfrac{g'}{2}\cos\theta_W \;=\; \tfrac{g_e}{2} \iff g'\cos\theta_W \;=\; g_e.$$

6.4 — The consistency relation (Griffiths Eq. 9.154)

Key Result

Griffiths Eq. 9.154 (the unification consistency relation):

$$\boxed{g_W \sin\theta_W \;=\; g'\cos\theta_W \;=\; g_e}$$

Solving for θ_W by dividing the two equalities:

$$\boxed{\tan\theta_W \;=\; \frac{g'}{g_W}}$$

Solving for the trig functions individually (using $\sin^2 + \cos^2 = 1$):

$$\sin\theta_W \;=\; \frac{g'}{\sqrt{g_W^2 + g'^2}}, \qquad \cos\theta_W \;=\; \frac{g_W}{\sqrt{g_W^2 + g'^2}}, \qquad g_e \;=\; \frac{g_W\, g'}{\sqrt{g_W^2 + g'^2}}.$$

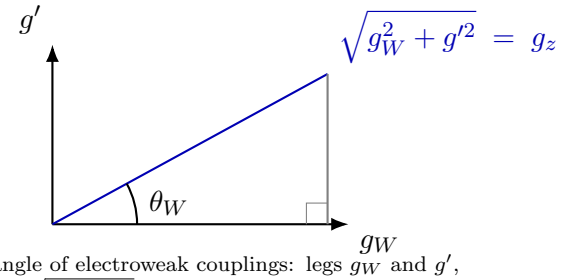
Griffiths quote

"The weak and electromagnetic coupling constants are not independent." — Griffiths §9.7.3, immediately following Eq. 9.154

Physical Insight

What this means. GWS theory has two free coupling constants (g_W and g') and one mixing angle (θ_W). The QED matching in Step 6.3 gives *two* constraints, relating these three quantities to the single electromagnetic coupling g_e. So in practice you can pick *any two* of $\{g_W, g', g_e, \theta_W\}$ as inputs and the others are determined. Griffiths puts it as: "the weak and electromagnetic coupling constants are not independent." This is what physicists mean when they say the electroweak theory "unifies" the two forces.

6.5 — Geometric picture: the right triangle of couplings



The triangle of electroweak couplings: legs g_W and g', hypotenuse $g_z = \sqrt{g_W^2 + g'^2} = g_W/\cos\theta_W$, slope $\tan\theta_W = g'/g_W$. Drop the perpendicular from the right angle to the hypotenuse to read off g_e.

6.6 — Experimental value (Griffiths Eq. 9.95 / 10.95)

Concept / Definition

The Weinberg angle is *not* predicted by GWS; it must be measured. Griffiths quotes (his Eq. 9.95 in the 2nd ed., Eq. 10.95 in the 1st ed.):

$$\sin^2\theta_W \approx 0.23, \qquad \theta_W \approx 28.7^\circ.$$

This puts the photon at $\sin^2\theta_W \approx 23\%$ admixture of W^3 inside B — not zero (which would mean $\gamma = B$ exactly and no unification), and not one (which would mean parity-violating QED). It's a non-trivial mixing.

Part 7 — The Z^0 coupling, and why does mixing happen?

Problem Statement

What this part does. (a) Read off the Z^0 coupling from Step 6.1 by using the consistency relation Eq. 9.154. This gives Griffiths Eqs. 9.155–9.156, which power Table 9.1 of c_V, c_A values. (b) Quote Griffiths' deferral of the "why" question to his §10.9 (the Higgs mechanism), and give a one-paragraph summary.

7.1 — The Z^0 coupling (Griffiths Eqs. 9.155–9.156)

Step 7.1: Use Eq. 9.154 to clean up the Z piece

From Step 6.1, the Z^μ coefficient in the rotated Lagrangian is

$$\mathcal{L}_Z \;=\; -\,i\,Z^\mu\Big(g_W\cos\theta_W\,j^3_\mu - \tfrac{g'}{2}\sin\theta_W\,j^Y_\mu\Big)\,.$$

Using $g_W = g_e/\sin\theta_W$ and $g' = g_e/\cos\theta_W$ (from Eq. 9.154):

$$\begin{aligned} g_W\cos\theta_W\,j^3 - \tfrac{g'}{2}\sin\theta_W\,j^Y &= \frac{g_e\cos\theta_W}{\sin\theta_W}\,j^3 - \frac{g_e\sin\theta_W}{2\cos\theta_W}\,j^Y \\ &= \frac{g_e}{\sin\theta_W\cos\theta_W}\big[\cos^2\theta_W\,j^3 - \tfrac{1}{2}\sin^2\theta_W\,j^Y\big]\,. \end{aligned}$$

Now use $j^Y = 2(j^{\rm em} - j^3)$ to eliminate j^Y:

$$\cos^2\theta_W\,j^3 - \tfrac{1}{2}\sin^2\theta_W\cdot 2(j^{\rm em}-j^3) \;=\; (\cos^2\theta_W + \sin^2\theta_W)\,j^3 - \sin^2\theta_W\,j^{\rm em} \;=\; j^3 - \sin^2\theta_W\,j^{\rm em}.$$

Therefore:

$$\boxed{\mathcal{L}_Z \;=\; -\,i\,g_z\,Z^\mu\big(j^3_\mu - \sin^2\theta_W\,j^{\rm em}_\mu\big)} \qquad \textbf{(Griffiths Eq. 9.155)}$$

where the Z^0 coupling constant is

$$\boxed{g_z \;\equiv\; \frac{g_e}{\sin\theta_W\cos\theta_W}} \qquad \textbf{(Griffiths Eq. 9.156)}.$$

Physical Insight

Why this matters. Eq. 9.155 is what Griffiths uses to derive Table 9.1 (the c_V, c_A couplings of the Z^0 to each fermion species). Specifically, expanding $j^3 - \sin^2\theta_W j^{\rm em}$ for each fermion gives the entries in his table. For example, for the electron neutrino (where $T_3 = +1/2$ and $Q = 0$): $c_V^{(\nu)} = T_3 = +1/2$ and $c_A^{(\nu)} = T_3 = +1/2$, which matches Griffiths' Table 9.1.

7.2 — Why does the mixing happen? (Griffiths defers to §10.9)

Griffiths quote

"All this raises some obvious questions: by what mechanism is the underlying $SU(2)_L \otimes U(1)$ symmetry of the electroweak interactions 'broken'? Why do the B and W^3 states 'mix' to form the Z^0 and the photon? If weak and electromagnetic interactions are, deep down, both manifestations of a single electroweak force, how come the weak mediators ($W^\pm$ and Z^0) are so heavy, while the electromagnetic mediator (γ) is massless? **I'll address these questions in the next chapter.**"
— Griffiths §9.7.3, end of section

Concept / Definition

Griffiths §10.9 ("The Higgs Mechanism") in one paragraph. A complex scalar field ϕ with potential $V(\phi) = -\mu^2|\phi|^2 + \frac{1}{2}\lambda^2|\phi|^4$ ($\mu^2 > 0$) has its minimum at $|\phi| = v \neq 0$, breaking the gauge symmetry spontaneously. When this scalar carries weak isospin and weak hypercharge (Griffiths' Eq. 10.115 generalized to the doublet case), the resulting Lagrangian, when expanded about the vacuum, contains mass terms for three of the four gauge fields. Specifically, the Higgs vacuum is electrically neutral, so the gauge field that couples to electric charge Q is left massless (Griffiths' "massless gauge field A^μ"), while the three orthogonal combinations get mass — these are the $W^\pm$ and Z^0. The mixing angle θ_W is the angle that diagonalizes the resulting mass matrix; it is fixed by the ratio g'/g_W, which is what Eq. 9.154 told us. **The Higgs mechanism is the dynamical reason for the mixing in Eq. 9.151**, and is the subject of Griffiths Ch. 10 (particularly §10.8 "Spontaneous Symmetry Breaking" and §10.9 "The Higgs Mechanism").

Intuition

One-sentence summary of why mixing happens. The Higgs vacuum is electrically neutral, so the gauge field that couples to electric charge Q stays massless — and *that* is the photon. Since Q is a linear combination of T_3 and Y (by Gell-Mann-Nishijima), the corresponding massless gauge field is the linear combination $A_\mu = B_\mu \cos\theta_W + W^3_\mu \sin\theta_W$ (Eq. 9.151). The orthogonal combination gets mass — it is the Z^0.

Final Summary — the answer to H2 in one box

Key Result

Following Griffiths §9.7 entirely:
Gauge group (Griffiths §9.7.2): $SU(2)_L \otimes U(1)$.
Four "fundamental" gauge fields:

$$\underbrace{W_\mu^1,\, W_\mu^2,\, W_\mu^3}_{SU(2)_L \text{ triplet (coupling } g_W)} \qquad \text{and} \qquad \underbrace{B_\mu}_{U(1) \text{ singlet (coupling } g')} .$$

The GWS Lagrangian (Griffiths Eq. 9.145):

$$\mathcal{L}_{\text{int}}^{\text{GWS}} \;=\; -\,i\Big[g_W\,\vec{j}_\mu \cdot \vec{W}^\mu \;+\; \tfrac{g'}{2}\, j_\mu^Y B^\mu\Big] .$$

Four physical bosons:

$$\boxed{\begin{aligned} W_\mu^\pm &= \tfrac{1}{\sqrt{2}}\big(W_\mu^1 \pm i\, W_\mu^2\big) && \textbf{(Griffiths Eq. 9.148)} \\ A_\mu &= \;\; B_\mu \cos\theta_W \;+\; W_\mu^3 \sin\theta_W && \textbf{(Griffiths Eq. 9.151)} \\ Z_\mu &= -B_\mu \sin\theta_W \;+\; W_\mu^3 \cos\theta_W && \textbf{(Griffiths Eq. 9.151)} \end{aligned}}$$

Coupling consistency relation (**Griffiths Eq. 9.154**):

$$g_W \sin\theta_W \;=\; g' \cos\theta_W \;=\; g_e, \qquad \tan\theta_W \;=\; g'/g_W, \qquad g_z \;=\; \frac{g_e}{\sin\theta_W \cos\theta_W}.$$

Experimental (Griffiths Eq. 9.95): $\sin^2\theta_W \approx 0.23$, $\theta_W \approx 28.7°$.
Why does the mixing happen? The $SU(2)_L \otimes U(1)$ symmetry is spontaneously broken to $U(1)_{\text{em}}$ by the Higgs mechanism (Griffiths Ch. 10, §10.9). The mixing angle θ_W is the angle that diagonalizes the resulting (W^3, B) mass matrix; the photon is the massless eigenvector (corresponding to the unbroken $Q = T_3 + Y/2$ generator), and Z^0 is the orthogonal massive eigenvector.

One-line cheat sheet

$$W^\pm = \tfrac{1}{\sqrt{2}}(W^1 \pm iW^2), \quad \binom{A}{Z} = \begin{pmatrix} \cos\theta_W & \sin\theta_W \\ -\sin\theta_W & \cos\theta_W \end{pmatrix}\binom{B}{W^3}, \quad g_W \sin\theta_W = g' \cos\theta_W = g_e.$$